

Functions: Let A and B be non empty sets. A function from A to B , denoted by $f: A \rightarrow B$ is a rule that assigns every element of A to exactly one element of B

$$A \ni a \xrightarrow{\text{maps to}} f(a) \in B \quad (f, A, B)$$

Some examples

$$\textcircled{1} \quad f: \mathbb{R} \longrightarrow \mathbb{R}, \quad f(x) = x^2$$

$$\textcircled{2} \quad f: \mathbb{R}^2 \longrightarrow \mathbb{R} \quad f(x, y) = x + y$$



One - one function (injective function)

A function $f: A \rightarrow B$ is called an one-one function if no two different elements of A are assigned to the same element of B .

Example / nonexample

$$\textcircled{1} \quad f: \mathbb{R} \longrightarrow \mathbb{R}, \quad f(x) = x^2 \text{ is NOT one-one because } f(1) = 1 \text{ and } f(-1) = 1.$$

$$\textcircled{2} \quad f: \mathbb{R}^2 \longrightarrow \mathbb{R} \quad f(x, y) = x + y \text{ is NOT one-one because } f(1, -1) = 0 = f(-1, 1).$$

③ $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = 2x$ is one-one because if $x \neq y$ then $2x \neq 2y$, that is, $f(x) \neq f(y)$.

④ Let $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ be an invertible matrix. If $Ax = Ay$ then multiplying by A^{-1} gives $x = y$. Thus for $x \neq y$, we must have $Ax \neq Ay$.

$x \mapsto Ax$

If $f: A \rightarrow B$ is one-one, then $f(a) = f(b) \Rightarrow a = b$

Onto functions (Surjective functions)

A function $f: A \rightarrow B$ is said to be onto if for every $b \in B$ there is an $a \in A$ such that $f(a) = b$.

Examples / nonexamples

① $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(x) = x^2$ is not onto because $x^2 \geq 0$ for any $x \in \mathbb{R}$ and thus $f(x) \neq -1$ for any $x \in \mathbb{R}$.

② $f: \mathbb{R} \rightarrow \mathbb{R}$; $f(x) = 2x$ is onto because

for any $y \in \mathbb{R}$, $f\left(\frac{y}{2}\right) = 2 \cdot \frac{y}{2} = y$.

③ If $A: \mathbb{R}^n \rightarrow \mathbb{R}^n$ is an invertible matrix
then for any $y \in \mathbb{R}^n$, we have
 $A(A^{-1}y) = y$.
Thus the function is onto.

$$\beta \in \mathbb{R}$$
$$f\left(\frac{\alpha+\beta}{2}, \frac{\alpha-\beta}{2}\right)$$
$$= \alpha.$$

④ for $f: \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x,y) = x+y$ is onto
because for any $\alpha \in \mathbb{R}$, $f(0,\alpha) = 0+\alpha = \alpha$.

Bijjective functions: A function that is both one-one and onto
is called a bijective function.

Linear Transformations (generalization of a matrix)

Let V and W be F -vector spaces

A function $T: V \rightarrow W$ is called a linear transformation
if the following two conditions hold

① $T(x+y) = T(x) + T(y)$; $x, y \in V$

② $T(\alpha x) = \alpha T(x)$; $x \in V, \alpha \in F$.

Conditions ① and ② can be combined to a single condition

③ $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$

for all $x, y \in V$ and $\alpha, \beta \in F$. (H.W.)

Examples / nonexamples

① $f: \mathbb{R} \rightarrow \mathbb{R}; f(x) = x^2$ is not a linear transformation because $f(x+y) = (x+y)^2$ but $f(x) + f(y) = x^2 + y^2$.

② $f: \mathbb{R}^2 \rightarrow \mathbb{R}; f(x, y) = x + y$ is a linear transformation because

$$\begin{aligned} \textcircled{\text{LT1}} \quad f((x_1, y_1) + (x_2, y_2)) &= f(x_1 + x_2, y_1 + y_2) \\ &= x_1 + x_2 + y_1 + y_2 \\ &= x_1 + y_1 + x_2 + y_2 \\ &= f(x_1, y_1) + f(x_2, y_2) \end{aligned}$$

$$\begin{aligned} \textcircled{\text{LT2}} \quad f(\alpha(x, y)) &= f(\alpha x, \alpha y) \\ &= \alpha x + \alpha y \\ &= \alpha(x + y) \\ &= \alpha f(x, y). \end{aligned}$$

③ Let $A: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be any matrix
 $m \times n$ matrix $x \mapsto Ax$.
then A is a linear transformation.

Kernel of a Linear transformation: Let $T: V \rightarrow W$ be a linear transformation between F -vector spaces

$$\text{Ker}(T) := \{ v \in V : T(v) = 0 \}.$$

Remarks

① $\text{Ker}(T)$ is a subspace of V .

Proof

① Let $v_1, v_2 \in \text{Ker}(T)$. Then $T(v_1 + v_2) = T(v_1) + T(v_2) = 0 + 0 = 0$.

Therefore $v_1 + v_2 \in \text{Ker}(T)$.

② Obvious! because $\text{Ker}(T) \subseteq V$.

③ $T(0) = T(0+0) = T(0) + T(0)$ $T(0) = T(0) + T(0)$

Therefore $T(0) = 0$. So $0 \in \text{Ker } T$

④ For any $v \in \text{Ker}(T)$,

$$T(-v) = T(-1 \cdot v) = -1 \cdot T(v) = -1 \cdot 0 = 0$$

Therefore $-v \in \text{Ker}(T)$.

This proves $\text{Ker}(T)$ is an abelian group. (Subgroup of V)

For any $\alpha \in F$ and $v \in \text{Ker}(T)$,

$$T(\alpha v) = \alpha T(v) = \alpha \cdot 0 = 0.$$

Therefore $\alpha v \in \text{Ker}(T)$.

Hence $\text{Ker}(T)$ is a subspace of V .

(①, ..., ④) automatically holds $\text{Ker}(T) \subseteq V$.

Theorem Let $T: V \rightarrow W$ be a linear transformation.
 $\ker(T) = \{0\}$ if and only if T is one-one

Proof Let $\ker(T) = \{0\}$ and $x, y \in V$.

If $T(x) = T(y)$ then $T(x) - T(y) = 0$

$$\Rightarrow T(x-y) = 0 \Rightarrow x-y \in \ker(T) = \{0\}.$$

$$\text{So } x-y=0 \Rightarrow x=y.$$

This proves that T is one-one.

Now Suppose that T is one-one and $v \in \ker(T)$.

$$\text{Then, } T(v) = 0$$

$$\Rightarrow T(v) = T(0).$$

But T is one-one then implies $v = 0$.

$$\text{Thus } \ker T = \{0\}.$$

Image of a linear Transformation

Let $T: V \rightarrow W$ be a linear Transformation.

Then $\text{im}(T) = \{T(v) : v \in V\}$ is an F -Vector space.

(H.W.)

- Note that $\text{im}(T) = W$ is the same as saying T is onto

Theorem. Let $T: \mathbb{R}^n \longrightarrow \mathbb{R}^m$ be a linear transformation.

Then, there is an $m \times n$ matrix A such that

$$T(x) = Ax. \text{ for all } x \in \mathbb{R}^n$$

Proof

Let $e_i = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix}$ 1 in i^{th} place and 0 everywhere else and let $e_i \in \mathbb{R}^n$

$$T(e_i) = \begin{pmatrix} x_{1i} \\ x_{2i} \\ \vdots \\ x_{mi} \end{pmatrix}$$

Then for $A = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & & x_{2n} \\ \vdots & \vdots & & \vdots \\ x_{m1} & x_{m2} & & x_{mn} \end{pmatrix}$

Arrows from the matrix columns to $T(e_1)$, $T(e_2)$, and $T(e_n)$

$$\text{We have } T(e_i) = Ae_i.$$

Now, for any point $x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix}$, we have

$$x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$x = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n$$

$$\begin{aligned} \Rightarrow T(x) &= x_1 T(e_1) + x_2 T(e_2) + \cdots + x_n T(e_n) \\ &= x_1 Ae_1 + x_2 Ae_2 + \cdots + x_n Ae_n \\ &= A(x_1 e_1 + x_2 e_2 + \cdots + x_n e_n) \\ &= Ax. \end{aligned}$$

□

Thus a linear transformation between $\mathbb{R}^n$ and $\mathbb{R}^m$ is simply a matrix multiplication.

Proposition Let $T: V \rightarrow W$ be a linear transformation of F -Vector Spaces.

If $v_1, \dots, v_n$ are F -linearly independent element of V and T is one-one, then $T(v_1), \dots, T(v_n)$ are F -linearly independent element of W .

Proof Suppose that $\alpha_1 T(v_1) + \dots + \alpha_n T(v_n) = 0$ for $\alpha_1, \dots, \alpha_n \in F$.

$$\text{Then } T(\alpha_1 v_1 + \dots + \alpha_n v_n) = 0.$$

That is, $\alpha_1 v_1 + \dots + \alpha_n v_n \in \text{Ker}(T)$. Since T is one-one,

we know that $\text{Ker}(T) = \{0\}$. Thus $\alpha_1 v_1 + \dots + \alpha_n v_n = 0$.

This implies $\alpha_1 = 0, \alpha_2 = 0, \dots, \alpha_n = 0$. □

$$a + 0 = a \text{ for all } a \in V$$

$$\Rightarrow 0 + 0 = 0$$

$$T(0) = T(0+0) = T(0) + T(0)$$